

JOURNAL OF EDUCATIONAL MANAGEMENT AND SUPERVISION (JEMAS)



Published in the Department of Educational Administration and Planning,
Joseph Sarwuan Tarka University, Makurdi

Redefining University Education in the Age of Artificial Intelligence: Implications for Teaching, Curriculum, and Postgraduate Supervision in Nigeria

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Abstract

This study investigated the implications of Artificial Intelligence (AI) for university education in Nigeria, with a specific focus on teaching practice, curriculum transformation, and postgraduate supervision. A descriptive survey design was employed, with a sample of 384 lecturers and postgraduate students selected from Nigerian universities using a multi-stage sampling procedure. Data were collected using a structured questionnaire and analyzed using Mean, Standard Deviation, and regression analysis. Findings revealed that AI significantly enhances teaching practice, curriculum transformation, and postgraduate supervision through improved instructional delivery, personalized learning, and efficient assessment systems. However, it also contributes to reducing critical thinking among students. The study further found that AI influences curriculum transformation by promoting digital and interdisciplinary competencies; however, such changes remain limited and largely superficial. In postgraduate supervision, AI was found to improve research productivity and academic writing, while simultaneously raising concerns regarding originality, authorship, and academic integrity. The study concludes that AI is a transformative yet paradoxical force in Nigerian higher education, requiring human-centered and ethically grounded integration. It recommends comprehensive policy frameworks, curriculum reform, and capacity building to ensure effective and responsible AI adoption.

Keywords: Artificial Intelligence, Curriculum Transformation, Teaching Practices, University Education, Postgraduate Supervision

Introduction

The advent of Artificial Intelligence (AI) is redefining the foundations of university education globally, with significant implications for teaching, curriculum design, and postgraduate supervision. Emerging scholarship suggests that AI-powered technologies are fundamentally reshaping higher education through adaptive learning systems, generative AI, and intelligent tutoring platforms that redefine instructional delivery and assessment practices (Long et al., 2026; Tan et al., 2025). Similarly, Iorhemen, Wasagu, and Dahiru (2026) affirm that the rapid advancement of AI has fundamentally altered how individuals access information, communicate, and solve problems in the 21st century. AI systems can analyze instructional performance, recommend resources, and personalize feedback based on evidence-driven instructional strategies, thereby improving educational quality and student outcomes (Almasri, 2024). However, the integration of AI into university systems presents a complex paradox. While AI enhances instructional delivery and research productivity, it simultaneously raises concerns about academic integrity, epistemic dependency, and the erosion of critical thinking (Peters & Olojede, 2025). Within the Nigerian and broader African context, these transformations are further complicated by structural constraints such as limited digital infrastructure, inadequate policy frameworks, and disparities in access to technological resources (Bali et al., 2024; Eleje et al., 2025). Despite growing interest in AI adoption, universities often lack coherent strategies for integrating AI into teaching, curriculum, and supervision practices.

Consequently, there is an urgent need to redesign university education in a manner that is context-sensitive, ethically grounded, and pedagogically sound. This study, therefore, examines the implications of AI for teaching, curriculum transformation, and postgraduate supervision within Nigerian universities.

Artificial Intelligence has been massively integrated into teaching, increasingly reshaping pedagogical practices globally and within African universities. AI tools such as intelligent tutoring systems, automated grading systems, and generative AI platforms enhance instructional delivery, improve feedback mechanisms, and promote personalized learning (Agbarakwe & Chibueze, 2024; Maluleke, 2025; Iorhemen et al, 2025). In the contexts of Nigeria, AI has been significantly influential in addressing challenges such as large class sizes and delayed assessment feedback. While AI enhances instructional efficiency and personalization (Agbarakwe & Chibueze, 2024), emerging scholarship warns that such gains may come at the cost of epistemic depth, as students increasingly outsource cognitive processes to algorithmic systems (Peters & Olojede, 2025). This tension reflects a broader debate between technological augmentation and intellectual autonomy in higher education. Overall, the literature reveals a dualistic trajectory in which AI simultaneously enhances pedagogical efficiency while undermining epistemic depth.

The transformation introduced by AI is driving curriculum reformation toward digital literacy, interdisciplinary learning, and problem-solving competencies in the 21st century and the reform is gradually integrated into Nigerian universities. Similarly, a study noted that African universities are gradually integrating AI into curricula, although progress remains uneven (Oguntona & Emere, 2025). Nevertheless, in Nigeria, curriculum reform is still at an early stage, often characterized by **surface-level integration of AI tools without deep epistemological restructuring** (Bali et al., 2024).

AI tools are increasingly integrated into postgraduate supervision for literature review synthesis, academic writing support and data analysis. These tools facilitate research productivity and provide additional support for students in resource-constrained environments (Obi & Ojo, 2025). Despite these benefits, AI introduces risks to originality and authorship, ethical dilemmas in assessing student work and over-reliance on AI for scholarly thinking. In Nigeria, the absence of formal guidelines worsens these challenges thereby creating doubts about the positive influence of AI on postgraduate supervision. The literature confirms that AI significantly reshapes supervision practices, but introduces **ethical and epistemological tensions**.

AI adoption in higher education raises critical ethical concerns, including academic integrity violations, data privacy and surveillance and algorithmic bias. In Nigeria and across Africa, these challenges are intensified by weak regulatory frameworks and limited institutional readiness (Afolabi, 2024). Contrary, AI is seen as democratizing access to education yet it may reinforce inequality due to unequal access to digital infrastructure. The literature strongly indicates that AI has **significant ethical and institutional implications**, particularly in developing contexts.

AI adoption in African universities is growing but remains uneven. Studies reveal increasing awareness and experimentation, limited large-scale institutional implementation and significant infrastructural and funding constraints (Eleje et al., 2025).

The reviewed literature reveals specific gaps thus: **Empirical gap:** there are limited quantitative studies testing relationships between AI and teaching, curriculum, and supervision in Nigeria. **Theoretical gap:** there is lack of integrated frameworks linking AI with pedagogical and supervisory theories. **Policy gap:** there are weak institutional and national AI governance structures while **contextual gap show over-reliance** on Global North literature. Finally, **supervision gap depict** absence of validated models for AI-assisted postgraduate supervision. By implication, the current study is justified as it addresses **under-researched variables in African contexts**, it contributes by **empirically testing hypotheses** rather than relying on conceptual claims and it bridges the gap between **theory, policy, and practice**.

Statement of the Problem

The rapid adoption of Artificial Intelligence in higher education has generated significant scholarly and policy interest. Nevertheless, its implementation remains uneven and theoretically underdeveloped, particularly in developing countries such as Nigeria. While existing studies highlight the potential of AI to enhance teaching efficiency, personalize learning, and support research activities, there is limited empirical evidence on how these transformations affect core academic processes, including pedagogy, curriculum structure, and supervision practices (Maluleke, 2025). Moreover, the emergence of generative AI tools has introduced new challenges related to academic integrity, originality of research, and the evolving role of the lecturer as a knowledge authority (Peters & Olojede, 2025). In many Nigerian universities, the absence of clear institutional policies and ethical guidelines has resulted in the unregulated use of AI technologies, raising concerns about quality assurance and educational standards (Afolabi, 2024).

Despite these developments, there remains a paucity of context-specific studies that systematically examine the combined implications of AI for teaching, curriculum, and supervision within Nigerian universities. This gap limits the development of evidence-based policies and pedagogical frameworks necessary for effective AI integration.

Objectives of the Study

The main objective of this study is to examine the implications of Artificial Intelligence for university education in Nigeria.

The specific objectives are to:

1. examine the influence of Artificial Intelligence on teaching practice in Nigerian universities.
2. assess the influence of Artificial Intelligence on curriculum transformation in Nigerian universities.
3. determine the influence of Artificial Intelligence on postgraduate supervision in Nigerian universities.

Research Questions

The study is guided by the following research questions:

1. How does Artificial Intelligence influence teaching practices in Nigerian universities?
2. What is the influence of Artificial Intelligence on curriculum transformation in Nigerian universities?
3. How does Artificial Intelligence affect postgraduate supervision in Nigerian universities?

Research Hypotheses

The following null hypotheses were tested:

1. Artificial Intelligence has no significant influence on teaching practices in Nigerian universities.
2. Artificial Intelligence has no significant influence on curriculum transformation in Nigerian universities.
3. Artificial Intelligence has no significant influence on postgraduate supervision in Nigerian universities.

Methodology

This study employed a quantitative descriptive survey design to examine the implications of Artificial Intelligence (AI) for teaching practice, curriculum transformation, and postgraduate supervision in Nigerian universities. The design was considered appropriate due to its capacity to collect standardized data from a large population and to describe relationships among variables. The study was conducted across selected federal, state, and private universities in Nigeria to ensure broad institutional representation. The selected universities included: Federal University of Kashere, Gombe State (Federal University), University of Jos, Plateau State (Federal University), Benue State University, Makurdi (State University), Federal University of Lafia (Federal University), Plateau State University, Bokkos (State University), Bingham University, Karu (Private University). These institutions were selected to reflect diversity in ownership structure, geographical spread (North-Central and North-East Nigeria), and varying levels of technological adoption.

The population of the study comprised all lecturers and postgraduate students in the selected universities. These groups were considered appropriate because they are directly involved in teaching, curriculum implementation, and research supervision processes. A sample size of 384 respondents was determined using Cochran's (1977) sample size determination formula for large populations: $n_0 = (Z^2pq) / e^2$.

Where: n_0 = initial sample size

- ✓ Z = standard normal deviation at 95% confidence level (1.96)
- ✓ p = estimated proportion of the population possessing the attribute (0.5 used for maximum variability)
- ✓ $q = 1 - p$ (0.5)

✓ e = margin of error (0.05)

Substituting the values: $n_0 = (1.96^2 \times 0.5 \times 0.5) / (0.05^2) \approx 384$. A multi-stage sampling procedure was then employed. First, stratified sampling was used to categorize institutions into federal, state, and private universities. Second, purposive sampling was applied to select faculties and departments where AI tools (e.g., learning management systems, automated grading tools, and generative AI platforms) are actively utilized. Finally, simple random sampling was used to select individual participants within each selected department. This procedure ensured representativeness, diversity, and inclusion of respondents with relevant exposure to AI-driven academic practices. Data were collected using a structured questionnaire titled *Artificial Intelligence and University Education Questionnaire (AIUEQ)*. The instrument was designed to measure four constructs thus: AI and Teaching Practices, AI and Curriculum Transformation, AI and Postgraduate Supervision and Demographic Characteristics. Responses were measured using a four-point Likert modified scale ranging from Strongly Agree (4) to Strongly Disagree (1). The instrument was subjected to face and content validation by experts in educational technology, measurement and evaluation, and higher education studies. Their feedback ensured clarity, relevance, and alignment with the study objectives. A pilot study was conducted using respondents outside the selected institutions. Reliability was established using Cronbach's Alpha coefficients were .82 for Teaching Practice, .79 for Curriculum Transformation, and .85 for Postgraduate Supervision indicating acceptable internal consistency.

Data were collected through both physical and online administration of the questionnaire. Hard copies were distributed within selected universities, while an online version was administered via Google Forms to reach participants across locations. This dual approach enhanced response rate and inclusivity. Data were analyzed using both descriptive and inferential statistical techniques. Descriptive statistics (Mean and Standard Deviation) were used to answer research questions, with a criterion mean of 2.50 as the decision threshold. Inferential statistics were conducted using simple linear regression analysis at a 0.05 level of significance and the regression model indicates a strong predictive relationship.

Ethical considerations were strictly observed throughout the study. Participation was voluntary, and informed consent was obtained from all respondents. Participants were assured of confidentiality and anonymity, and data collected were used solely for academic purposes.

Results

This section presents the analysis of data collected from 384 respondents, comprising lecturers and postgraduate students from selected universities in Nigeria. Data are presented using descriptive statistics (mean and standard deviation) and inferential statistics (t-test and regression analysis) in line with the research questions and hypotheses.

Demographic Characteristics of Respondents

Table 1: Distribution of Respondents by Category

Category	Frequency (n)	Percentage (%)
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Lecturers	170	44.2
Postgraduate Students	214	56.0
Total	384	100

Table 1 depicts that **56% of respondents are postgraduate students**, while **44.2% are lecturers**. This indicates a balanced representation of both groups, ensuring that perspectives on teaching, curriculum, and supervision are adequately captured.

Research Question One: How does AI influence teaching practice in Nigerian universities?

Table 2: Mean Responses on AI and Teaching Practice (n = 384)

S/N	Item	Mean	SD	Decision
1.	AI improves instructional delivery	3.21	0.76	Agree
2.	AI enhances feedback and assessment efficiency	3.34	0.71	Agree
3.	AI supports personalized learning	3.18	0.80	Agree
4.	AI reduces lecturers' workload	3.05	0.85	Agree
5.	AI negatively affects students' critical thinking	2.87	0.76	Agree
	Grand Mean	3.13		Agree

Table 2 shows the grand mean of **3.13** which indicate those respondents agree that AI substantially influences teaching practice. While AI enhances efficiency and personalization, respondents also acknowledge its **negative influence on critical thinking**, highlighting a dual influence.

Research Question Two: What is the influence of AI on curriculum transformation?

Table 3: Mean Responses on AI and Curriculum Transformation

S/N	Item	Mean	SD	Decision
1.	AI promotes integration of digital skills into curriculum	3.26	0.74	Agree
2.	AI encourages interdisciplinary learning	3.11	0.79	Agree
3.	Curriculum reflects AI-related competencies	2.68	0.79	Agree
4.	AI has led to major curriculum restructuring	2.41	0.92	Disagree
5.	Universities are adequately updating curriculum for AI	2.36	0.95	Disagree
	Grand Mean	2.76		Agree

Table 3 shows a grand mean of **2.76** suggesting that AI influences curriculum transformation; however, the low mean scores on curriculum restructuring show that **changes are still limited and superficial** in Nigerian universities.

Research Question Three

How does AI influence postgraduate supervision?

Table 4: Mean Responses on AI and Postgraduate Supervision

S/N	Item	Mean	SD	Decision
1.	AI assists in literature review and research writing	3.38	0.69	Agree
2.	AI improves research productivity	3.29	0.73	Agree

3. AI aids communication between supervisor and students	3.95	0.84	Agree
4. AI threatens originality in research	3.12	0.88	Agree
5. AI creates ethical challenges in supervision	3.25	0.77	Agree
Grand Mean	3.20		Agree

Table 4 depicts the grand mean of **3.20** indicating that AI considerably influences postgraduate supervision. While it aids research efficiency, it also introduces **serious concerns regarding originality and ethics**.

Hypotheses Testing

Hypothesis One (H01): AI has no significant influence on teaching practice

Table 5: Regression Analysis of AI and Teaching Practice

Variable	B	SE	Beta	t	p-value
AI Influence	0.62	0.08	0.58	7.75	.000

Table 5 indicates that since $P < .05$, the null hypothesis is rejected. This shows that AI has a significant positive influence on teaching practices in Nigerian universities.

Hypothesis Two (H02): AI has no significant influence on curriculum transformation

Table 6: Regression Analysis of AI and Curriculum Transformation

Variable	B	SE	Beta	t	p-value
AI Influence	0.44	0.09	0.41	4.89	.000

Table 6 shows that since $p < .05$, the null hypothesis is rejected. AI profoundly influences curriculum transformation; however the effect size is **moderate**, showing partial adoption.

Hypothesis Three (H03): AI has no significant influence on postgraduate supervision

Table 7: Regression Analysis of AI and Supervision

Variable	B	SE	Beta	t	p-value
AI Influence	0.67	0.07	0.63	9.12	.000

Table 7 indicates that since $p < .05$, the null hypothesis is rejected. This shows that AI has a **strong and significant influence on postgraduate supervision**, particularly in research support.

Discussion of the Findings

The findings of this study validate the assumption that AI is significantly reshaping university education in Nigeria across **teaching, curriculum, and postgraduate supervision**, albeit in complex and sometimes a paradoxical pattern. First, the study found that AI enhances teaching practice through reformed instructional delivery, personalized learning, and efficient assessment. The findings extend existing scholarship by empirically confirming the dualistic nature of AI in higher education, where efficiency gains coexist with cognitive trade-offs. This aligns with

emerging global concerns regarding ‘algorithmic dependency (Long et al., 2025; Tan et al., 2025). Contrary, the finding that AI may weaken students’ critical thinking supports concerns that excessive reliance on generative AI can lead to **cognitive dependency and superficial learning** (Peters & Olojede, 2025). This paradoxical pattern reflects a central tension in AI adoption efficiency versus intellectual rigor. Second, the study demonstrated that AI influences curriculum transformation by promoting digital and interdisciplinary competencies; however that such transformation remains limited and largely peripheral. This corroborates with earlier studies that highlight **slow and uneven curriculum reform** in African universities due to institutional inactivity and infrastructural constraints (Bali et al., 2024; Maluleke, 2025). By implication, while AI is driving change, it is yet to fundamentally transform curriculum structures. Third, the study established that AI significantly enhances postgraduate supervision by supporting research processes such as literature review and academic writing. This is in line with emerging literature positioning AI as a research support tool (Obi & Ojo, 2025). However, concerns about originality and academic integrity raises wider debates about authorship and ethical use of AI in scholarly work. From a practical perspective, the study underscores the need for **human-centered AI integration** in Nigerian universities. It highlights the importance of training lecturers in AI-informed pedagogy, reforming curricula to include AI literacy and ethics and implementing clear guidelines for AI use in research and supervision. Overall, the study contributes empirical evidence that AI is not merely a technological tool but a **transformative force that requires meticulous pedagogical, institutional, and ethical positioning**.

Conclusion

This study concludes that Artificial Intelligence is reshaping university education in Nigeria by transforming teaching practice, curriculum development, and postgraduate supervision. AI enhances instructional efficiency, supports personalized learning, and improves research productivity; however, it also raises concerns regarding critical thinking, academic integrity, and over-reliance on automated systems. Although Nigerian universities are gradually integrating AI into academic processes, curriculum transformation and institutional preparedness remain limited. Therefore, effective AI integration requires deliberate pedagogical alignment, ethical regulation, and sustained institutional support to ensure that AI complements rather than replaces human intellectual engagement in higher education.

Recommendations

The following recommendations are made based on the results of this study.

1. **Curriculum Reform:** Universities should integrate AI literacy, digital competencies, and ethical reasoning into undergraduate and postgraduate curricula across disciplines.
2. **Capacity Building:** Continuous professional development programmes should be organized to equip lecturers and supervisors with AI-related pedagogical and research competencies.
3. **Ethical Guidelines:** Universities and regulatory bodies should establish clear policies governing the ethical use of AI in teaching, assessment, and postgraduate research supervision.

4. **Promotion of Critical Thinking:** Lecturers should adopt human-AI collaborative approaches that encourage analytical thinking, creativity, and independent scholarship rather than passive dependence on AI-generated outputs.
5. **Institutional Support:** Government and university authorities should strengthen digital infrastructure, funding, and AI governance frameworks to support equitable and sustainable AI adoption in Nigerian higher education institutions.

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